TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
Intercepts: set y = 0 to find the x-intercept; set x = 0 to find the y-intercept.
Solution of a system: the point where the lines cross. It makes both equations true.
How many? Different slopes: one. Same slope, different y-intercept: none. Same line: infinite.

1. Which point is the solution of $y = 2x$ and $y = x + 3$?

Tip: the solution makes both equations true

A. (3, 6)

B. (6, 3)

C. (1, 2)

2. The graphs of $y = 3x + 1$ and $y = 3x - 4$ are…

Tip: same slope, different b: parallel

F. lines that cross once

G. parallel lines

H. the same line

3. Which point is on the line $5s + 10a = 3000$?

> **Tip:** substitute and check

A. (300, 200)

B. (100, 300)

C. (200, 200)

4. What are the slope and y-intercept of $y = 4x + 40$?

> **Tip:** m = slope, b = y-intercept

F. slope 40, y-intercept 4

G. slope 4, y-intercept -40

H. slope 4, y-intercept 40